Artificial Intelligence for Antimicrobial Resistance (AMR) Surveillance

Research Paper and Platform Design — RM A1 Assignment

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PART A: RESEARCH PAPER

Abstract

Background: Antimicrobial resistance (AMR) represents one of the most pressing public health challenges of this century. The WHO has classified AMR as a global health priority, with projections indicating that drug-resistant infections could cause 10 million deaths annually by 2050 without decisive action (WHO, 2023). The convergence of irrational antibiotic prescribing, declining pharmaceutical innovation pipelines, substandard drug formulations, and fragmented global surveillance has deepened the crisis. Existing surveillance mechanisms are geographically inconsistent and often delayed.

Objective: This study examines how artificial intelligence (AI) can be applied as a multi-domain surveillance and intervention tool in AMR management. Specifically, it investigates: (a) how AI-powered prescriber behaviour analytics can reduce inappropriate antibiotic use; (b) the relationship between drug formulation stability and resistance emergence; (c) AI-driven optimisation of combination therapy for resistant pathogens; and (d) computational repurposing of existing compounds for AMR applications.

Methodology: A sequential explanatory mixed-methods design integrating multivariate regression, predictive modelling, and thematic analysis of healthcare professional (HCP) interviews. Secondary data from WHO GLASS, CDDEP, and PubChem databases. Primary data collection targeting 250 HCPs across tertiary and community healthcare settings.

Practical Implications: The research underpins the design of the AMR-Genius Intelligence Hub — a proposed AI-powered surveillance platform integrating four analytical modules: HCP Action Intelligence, Formulation Stability Analytics, Combination Therapy Optimizer, and Compound Repurposing Engine.

Keywords: *antimicrobial resistance; AI surveillance; machine learning; prescriber behaviour; drug repurposing; combination therapy; pharmaceutical management*

1. Introduction

1.1 Problem Statement

Antimicrobial resistance occurs when bacteria, viruses, fungi, and parasites evolve mechanisms to resist the drugs designed to eliminate them, rendering standard treatments increasingly ineffective. In 2019 alone, bacterial AMR was directly responsible for 1.27 million deaths globally, with attributable mortality rising sharply in low- and middle-income countries (Murray et al., 2022). The World Bank projects that AMR could reduce global GDP by up to 3.8% by 2050, with associated healthcare costs escalating by hundreds of billions of dollars annually (World Bank, 2017).

Several interconnected drivers sustain the crisis. Over-prescription and misuse of antibiotics by healthcare professionals — driven by diagnostic uncertainty, patient demand, and systemic incentive structures — remains the most preventable. Studies indicate that 30–50% of antibiotic prescriptions in hospital settings are either unnecessary or inappropriate (CDC, 2022). Separately, poor-quality antimicrobial formulations — characterised by subtherapeutic drug concentrations — expose bacterial populations to non-lethal antibiotic levels and accelerate resistance selection (Almuzaini et al., 2013). The pharmaceutical industry has largely withdrawn from antibiotic R&D; due to unfavourable return profiles, and surveillance systems lack the real-time responsiveness needed to guide policy effectively.

1.2 Rationale for AI Integration

Machine learning, deep learning, and natural language processing offer capabilities that conventional methods cannot match: processing large, heterogeneous datasets from clinical, genomic, environmental, and pharmacological sources; identifying non-linear patterns predictive of resistance emergence; optimising drug combinations for synergistic effect; and screening large chemical libraries for repurposable compounds — all at a speed and scale beyond manual analysis (Stokes et al., 2020).

The critical observation underlying this study is that despite the availability of powerful AI tools and rich surveillance databases (WHO GLASS, CDDEP, NCBI GenBank, PubChem), the pharmaceutical management sector has been slow to integrate these into cohesive, deployable platforms. This research bridges the gap between technical potential and strategic application, with direct implications for pharmaceutical companies, regulatory agencies, hospital stewardship programmes, and public health policymakers.

1.3 Scope

The study addresses AI applications across four domains: prescriber behaviour analytics, drug quality surveillance, therapeutic optimisation, and compound repurposing. Part A provides the literature review, methodology, and managerial implications. Part B extends the findings into the design of the AMR-Genius Intelligence Hub digital platform.

2. Literature Review

2.1 AI in AMR Surveillance: Global Landscape

Foundational work by Stokes et al. (2020), published in Cell, demonstrated that a deep learning model could identify halicin — a structurally novel antibiotic compound — from a chemical library of over 100 million molecules with demonstrated efficacy against extensively drug-resistant Mycobacterium tuberculosis and pan-resistant Acinetobacter baumannii. This landmark study reframed AI's role from passive surveillance to active therapeutic innovation.

Melo et al. (2021) applied Random Forest and Gradient Boosting algorithms to predict antimicrobial susceptibility profiles from whole-genome sequencing data with accuracy rates exceeding 95% for Klebsiella pneumoniae and Escherichia coli. AI-driven susceptibility prediction from genomic data could replace or supplement phenotypic culture-based methods that typically require 48–72 hours. Liu et al. (2020) demonstrated that NLP applied to electronic health records could identify inappropriate antibiotic prescriptions with 87% sensitivity — a significant proof point for AI-enabled stewardship programmes.

2.2 HCP Prescriber Behaviour and AI

Inappropriate prescribing remains the most modifiable driver of AMR. Gerber et al. (2014) showed in a randomised controlled trial that prescriber performance feedback significantly reduced inappropriate antibiotic prescribing in outpatient paediatric settings, confirming that clinicians respond to data-driven interventions. Meeker et al. (2016) extended this, demonstrating that accountable justification, peer comparison, and suggested alternatives — all amenable to AI implementation — reduced inappropriate prescribing by 33% in ambulatory care.

More recent work by Peiffer-Smadja et al. (2020) found that while AI-powered clinical decision support systems for antibiotic selection show demonstrable technical efficacy, adoption barriers — clinician trust, workflow friction, and concerns about algorithmic transparency — remain significant. These findings underscore that successful deployment is a management and change-management challenge as much as a technical one.

2.3 Drug Formulation Quality and AMR

The relationship between substandard antimicrobial formulations and resistance is mechanistically established but operationally underexplored in the surveillance literature. Almuzaini et al. (2013) estimated that substandard and falsified medicines constitute approximately 10.5% of medicines in low- and middle-income countries. Subtherapeutic concentrations allow bacterial populations to persist and develop resistance through natural selection, effectively accelerating the mutation-selection cycle.

Temple et al. (2021) proposed a predictive AI framework integrating supply chain temperature monitoring data, storage condition logs, and dissolution testing results to forecast formulation degradation and flag at-risk drug batches before distribution. This framework has not been applied at scale within an integrated AMR surveillance context — one of the specific gaps this research addresses through the Formulation Stability module.

2.4 Combination Therapy Optimisation

For a panel of 20 antibiotics, there are 190 pairwise combinations and 1,140 triple combinations — rendering exhaustive experimental screening impractical. Menden et al. (2019) developed a neural network model that predicted synergistic drug combinations with a Pearson correlation of 0.73 against experimental observations, a methodology directly transferable to antimicrobial applications. Baym et al. (2016) showed through experimental evolution that certain drug combinations impose evolutionary constraints that severely limit resistance development pathways — suggesting AI-optimised combinations can be designed not just for efficacy but for evolutionary robustness.

2.5 AI-Driven Drug Repurposing

Repurposing existing approved compounds for antimicrobial applications offers a strategically attractive route to expanding the antibiotic armamentarium at reduced development timelines and cost. Mikatis et al. (2022) demonstrated that graph neural network models applied to protein-ligand interaction databases could identify 14 non-antibiotic compounds with significant antibacterial activity against MRSA, including compounds with synergistic effects when combined with conventional antibiotics at sub-minimum inhibitory concentrations.

2.6 Research Gap

Despite substantial evidence supporting AI's potential in individual AMR-related domains, no integrated, commercially deployable platform currently unifies prescriber behaviour analytics, formulation quality surveillance, combination therapy optimisation, and drug repurposing within a single management-accessible interface. Existing platforms are domain-specific and technically inaccessible to pharmaceutical management decision-makers. No published study has systematically assessed the managerial feasibility, revenue model viability, and operational integration requirements of such a platform — the gap this research addresses.

3. Research Methodology

3.1 Research Philosophy and Design

A pragmatist research philosophy is adopted, acknowledging that method selection should be guided by practical imperatives rather than adherence to a single epistemological tradition. A sequential explanatory mixed-methods design is proposed: Phase 1 (quantitative) conducts large-scale secondary data analysis and regression modelling; Phase 2 (qualitative) applies thematic analysis of semi-structured HCP interviews to contextualise the quantitative findings. Integration occurs at the interpretation stage.

3.2 Secondary Data Sources

Database	Relevance	Access	Module
WHO GLASS (glass.who.int)	National AMR prevalence, antibiotic consumption data, resistance trends by pathogen-drug combination	Public download	HCP Action Intelligence, Formulation Stability
CDDEP ResistanceMap	Country-level resistance rates, antibiotic use metrics, interactive surveillance data	Public; API-supported	HCP Action Intelligence, Combination Therapy
NCBI GenBank / Pathogen Portal	Whole-genome sequencing data for resistant isolates; genomic resistance determinants	Public; Entrez API	Combination Therapy, Repurposing
PubChem Compound Database (NIH)	Chemical structures, bioactivity data, pharmacological properties for repurposing candidates	Public; RESTful API	Repurposing Engine
EUCAST / CLSI MIC Databases	Minimum inhibitory concentration breakpoints; susceptibility testing standards	Public	Combination Therapy
FDA FAERS / EudraVigilance	Post-market drug safety data; formulation quality signals	Public quarterly download	Formulation Stability

Database	Relevance	Access	Module
ICMR AMR Surveillance Network	Geographically relevant national resistance data for Indian healthcare context	Public reports	All modules

3.3 Primary Data Collection

The primary data collection targets healthcare professionals involved in antimicrobial prescribing and dispensing. The target population is stratified into: hospital physicians and intensivists at tertiary care centres, community GPs and primary care physicians, and clinical pharmacists and antimicrobial stewardship officers.

For the quantitative survey component, stratified random sampling is applied. The target sample is N = 250 HCPs: 100 hospital-based physicians (Stratum 1), 100 community practitioners (Stratum 2), and 50 clinical pharmacists and stewardship officers (Stratum 3). Sample size is based on a 95% confidence level, ±5% margin of error, and an assumed population proportion of 0.50, yielding a minimum n = 196; the target of 250 provides a 27.6% buffer for attrition.

For qualitative depth interviews, a purposive sample of 20–25 HCPs is drawn from the quantitative pool, ensuring variation across specialty, geographic setting, years of experience, and institutional affiliation. Data saturation serves as the terminal criterion for qualitative sampling (Guest et al., 2006).

3.4 Quantitative Analysis

The primary quantitative tool is multivariate linear regression (MLR), using AMR prevalence rates sourced from WHO GLASS as the dependent variable, with independent predictors including antibiotic consumption rates (defined daily doses per 1,000 inhabitants per day from CDDEP), prescriber adherence scores, drug quality indices, and healthcare infrastructure variables. All models are subjected to diagnostic checks for multicollinearity (VIF < 5), homoscedasticity (Breusch-Pagan test), and normality of residuals.

A Random Forest classifier is trained on the WHO GLASS dataset to identify country-level and institutional-level predictors of high AMR prevalence, enabling feature importance ranking that informs the platform modules. Model performance is evaluated by AUC-ROC, precision, recall, and F1-score using 10-fold cross-validation.

3.5 Qualitative Analysis

Thematic analysis follows the six-phase framework established by Braun and Clarke (2006). NVivo 14 is used to manage and systematically code qualitative data. Inter-rater reliability is established by having 20% of transcripts independently coded by a second researcher, targeting a Cohen's Kappa coefficient of ≥ 0.70. Member checking will validate interpretive accuracy with a sub-sample of interviewees (Lincoln and Guba, 1985).

3.6 Ethical Considerations

All primary data collection will comply with the ICMR National Ethical Guidelines for Biomedical and Health Research (2017). Written informed consent will be obtained from all participants. Anonymisation protocols ensure no individual HCP or institution is identifiable in research outputs. Institutional ethics committee approval will be sought prior to commencement of primary data collection.

3.7 Limitations

Self-report bias in the HCP questionnaire may lead to social desirability effects, with respondents potentially over-reporting guideline-concordant prescribing. The cross-sectional quantitative phase precludes causal inference. Secondary database quality depends on the completeness and consistency of national reporting to WHO GLASS, which varies significantly across countries. These limitations will be mitigated through triangulation across data sources and cautious interpretation of causal language.

4. Managerial Implications

4.1 Strategic Recommendations

Pharmaceutical management professionals should reframe AI-powered AMR surveillance as a competitive differentiator rather than a cost centre. Companies that invest early in integrated surveillance infrastructure — combining prescriber behaviour analytics and compound repurposing capabilities — will be better positioned to anticipate resistance trends, guide pipeline prioritisation, and build credibility with regulatory bodies increasingly demanding evidence-based antimicrobial stewardship commitments.

From a portfolio management perspective, the drug repurposing strand highlights the strategic value of mining existing compound libraries. Given development timelines of 10–15 years and costs exceeding USD 1 billion for new antibiotic approvals, AI-accelerated repurposing can deliver a 60–70% reduction in time-to-market for repurposed antimicrobial candidates — a compelling option for companies constrained by R&D; budgets (Pushpakom et al., 2019).

4.2 Operational Recommendations

Healthcare institutions should prioritise embedding AI-powered clinical decision support tools within existing EMR systems to minimise workflow disruption — the primary adoption barrier identified in the literature. The HCP Action Intelligence module should function as a real-time prescribing alert and peer-comparison feedback system delivered through the prescriber's existing interface rather than as a standalone application.

Quality assurance teams within pharmaceutical manufacturers should integrate the Formulation Stability module's predictive analytics into cold-chain monitoring workflows, enabling proactive batch recall decisions before substandard products reach end-users. The mechanistic evidence linking subtherapeutic drug concentrations to resistance selection makes this a patient safety priority, not only a commercial one.

4.3 Financial Case

Investment in AI-driven AMR surveillance has a strongly favourable cost-benefit profile. The CDC estimates that healthcare-associated infections caused by resistant organisms impose a direct economic burden of USD 35 billion annually in the United States alone. Modelling by Laxminarayan et al. (2016) suggests that a 30% reduction in inappropriate antibiotic prescribing — achievable through AI-enabled stewardship based on existing intervention evidence — could prevent between 37,000 and 99,000 AMR-related deaths annually in low- and middle-income countries, representing avoided productivity losses of USD 3–8 billion.

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PART B: AMR-GENIUS INTELLIGENCE HUB — PLATFORM DESIGN

5. Platform Overview and Architecture

The AMR-Genius Intelligence Hub is a proposed cloud-based, AI-powered surveillance and decision-support platform that unifies four critical AMR management domains within a single management-accessible interface. Its core value proposition is democratising advanced AMR intelligence: translating complex genomic, pharmacological, and epidemiological data streams into actionable real-time insights for pharmaceutical professionals, hospital stewardship teams, and public health policymakers — without requiring computational expertise from end-users.

The platform operates on a three-tier architecture: (1) a Data Ingestion and Processing Layer continuously pulling from WHO GLASS, CDDEP, NCBI, PubChem, and institutional EHR systems via standardised HL7 FHIR APIs; (2) an AI Analytics Engine Layer comprising the four ShinyApp modules hosted on cloud ML infrastructure (AWS SageMaker or Azure ML); and (3) a Dashboard and Reporting Interface Layer providing role-specific visualisations, alerts, and exportable reports through a responsive web application. The platform is designed to comply with HIPAA, GDPR, and India's Personal Data Protection Act.

6. Target Users and Business Problem

User Segment	Role in AMR Management	Primary Module	Decision Supported
Hospital CMOs & Antimicrobial Stewardship Officers	Institutional prescribing governance, policy	HCP Action Intelligence	Prescribing protocol updates, HCP feedback
Clinical Pharmacists	Drug selection, dosage optimisation	Combination Therapy Optimizer	Synergistic combination selection
Pharma QA / Regulatory Affairs Managers	Drug quality compliance, batch release	Formulation Stability Analytics	Proactive batch recall, distribution decisions
Pharma R&D; Pipeline Managers	Drug discovery strategy, portfolio planning	Repurposing Engine	Compound prioritisation for AMR indication
National Public Health Officials (e.g. ICMR)	National AMR surveillance, policy formulation	All modules (aggregate view)	AMR containment strategies, resource allocation

Unique Selling Points

Integration breadth: The only platform combining prescriber behaviour, formulation quality, combination therapy, and repurposing intelligence within a single interface — eliminating the need for multiple specialised tools.

Management accessibility: AI insights are translated into plain-language alerts, traffic-light risk ratings, and exportable PDF reports, making advanced analytics accessible to C-suite pharmaceutical managers without bioinformatics training.

Real-time responsiveness: Continuous data ingestion and automated model retraining ensure that resistance trend predictions and drug recommendations remain current, addressing the critical latency problem of traditional surveillance approaches.

7. Platform Modules: Features and Analytics

Module	Description	AI Technology	Business Value
HCP Action Intelligence	Real-time tracking and analysis of antibiotic prescribing patterns by individual HCPs, departments, and institutions. Generates peer-comparison benchmarking, flags outlier prescribers, delivers personalised stewardship alerts via EMR integration.	NLP on EHR data; Gradient Boosting classifiers; peer-comparison algorithm	Targets the most modifiable AMR driver. Evidence indicates 33% reduction in inappropriate prescriptions through peer feedback (Meeker et al., 2016).
Formulation Stability Analytics	Predictive monitoring of drug quality across supply chains. Ingests temperature logs, storage condition data, batch test results, and pharmacovigilance signals to forecast degradation risk and issue pre-emptive batch alerts.	LSTM neural networks; anomaly detection algorithms; supply chain graph analytics	Prevents subtherapeutic antibiotic distribution — a mechanistic driver of resistance. Reduces regulatory non-compliance risk.
Combination Therapy Optimizer	AI engine for identifying synergistic drug combinations against specific resistant pathogens. Takes pathogen identity and outputs ranked combination recommendations with predicted synergy scores and resistance-evasion profiles.	Neural network synergy prediction model (trained on EUCAST MIC data); graph neural networks for drug-drug interaction mapping	Extends therapeutic options against resistant organisms without new drug development. Supports prescribers in navigating combinatorial complexity.
Compound Repurposing Engine	AI-driven screening of PubChem and FDA-approved compound libraries for antimicrobial activity against priority resistant pathogens. Generates ranked candidates with predicted activity profiles, safety considerations, and clinical trial status.	Graph Neural Networks on molecular structure data; molecular docking simulation; knowledge graph querying	Accelerates antimicrobial pipeline at fraction of de novo development cost. Provides pharma R&D; teams with evidence-based repurposing leads.

8. Feasibility Analysis

Technical Feasibility

Technical feasibility is assessed as high. All component technologies — cloud ML infrastructure (AWS SageMaker, Azure ML), HL7 FHIR API integration, NLP text analytics, GNN molecular screening, and real-time IoT data ingestion — are commercially mature with established implementation precedents in healthcare. Open-source ML frameworks (TensorFlow, PyTorch, scikit-learn) reduce development costs significantly. Key technical risks are: HL7 FHIR integration complexity across heterogeneous hospital IT systems (mitigated by a phased approach); data quality variability from WHO GLASS (mitigated by imputation modelling); and model drift as resistance patterns evolve (mitigated by automated retraining pipelines).

Financial Feasibility and Revenue Model

Revenue Stream	Target Customer	Pricing	Est. Annual Revenue (Year 3)
Institutional SaaS — Standard (HCP Action Intelligence + Dashboard)	Hospital systems (150–500 beds)	Annual licence fee	USD 2.1M (70 hospitals × USD 30,000)
Institutional SaaS — Enterprise (full platform + EHR integration)	Large hospital groups, NHS-equivalent systems	Annual licence + implementation fee	USD 3.6M (24 enterprise × USD 150,000)
Government / Public Health Agency Licensing	ICMR, WHO regional offices, national health ministries	Annual licence	USD 1.2M (12 agencies × USD 100,000)
Pharma R&D; Intelligence Partnerships (Repurposing Engine)	Top 20 pharmaceutical companies	Milestone-based + royalty	USD 4.0M (4 partners × USD 1M)
De-identified Data Analytics Service	Pharma marketing & market access teams	Per-report or annual subscription	USD 800K (aggregate subscription)

Projected total revenue (Year 3): USD 11.7 million | Platform development cost estimate: USD 3.5–4.0M | Break-even: Month 26–30 at 55% subscription target achievement.

Operational Feasibility

Operational feasibility is assessed as moderate-to-high, contingent on securing two or three anchor hospital partners for early integration pilots. Primary operational challenges are: institutional data governance negotiations for EHR access (18–24 month procurement cycles typical); change management requirements for HCP adoption of prescribing alert systems; and maintaining a multi-disciplinary product team spanning clinical informatics, AMR science, ML engineering, and pharmaceutical regulatory expertise.

A phased go-to-market strategy is recommended: Phase 1 (Months 1–12) — deploy Formulation Stability and Repurposing modules as standalone tools for pharmaceutical R&D; clients (shorter procurement cycles); Phase 2 (Months 13–24) — onboard hospital pilot partners for HCP Action Intelligence; Phase 3 (Months 25–36) — full platform deployment with government health agency partnerships.

9. Platform Interface Design

Main Dashboard: Full-width header with user role tag and notification bell. A three-column KPI row shows the National AMR Risk Index (circular gauge), Inappropriate Prescriptions Today (real-time counter with trend sparkline), and Active Resistance Alerts (count badge). Main content is a 2×2 module grid; each cell shows module status and a Launch Module button. A collapsible sidebar provides navigation to Settings, Data Sources, Reports, and Help.

HCP Action Intelligence: Search/filter bar at top for hospital, department, specialty, or date range. Primary visualisation is a horizontal bar chart ranking prescribers by Inappropriate Prescribing Score (IPS), with the current user's bar highlighted and the departmental average marked by a dashed vertical line. A detail card on right shows total prescriptions, IPS breakdown by antibiotic class, top guideline deviations, and a personalised recommendation.

Formulation Stability: World map (Leaflet.js) as hero element with batch shipment locations as colour-coded dots (green: stable, amber: at-risk, red: critical degradation predicted). Timeline slider adjusts temporal view. Clicking any batch opens a detail drawer showing the full temperature log as a line chart with threshold violation zones shaded, and an AI-generated recommendation.

Combination Therapy Optimizer: Two-panel input-output design. Left panel: pathogen selection dropdown (WHO Priority Pathogen List), resistance profile checkboxes, and patient context parameters. Right panel: ranked drug combination table showing Drug A + Drug B, Predicted Synergy Score, Mechanism of Synergy, Resistance Evasion Rating, and Clinical Evidence Grade.

Compound Repurposing Engine: Chemical structure search panel with SMILES string or compound name input. Main area displays an interactive D3.js network graph where nodes are compounds and edges represent structural similarity or shared protein targets. Ranked candidate list below shows Compound Name, Current Indication, Target Pathogen, Predicted MIC Reduction (%), Patent Status, and Development Stage.